Design and Development of a Head-Tracking FPV Camera Control Board

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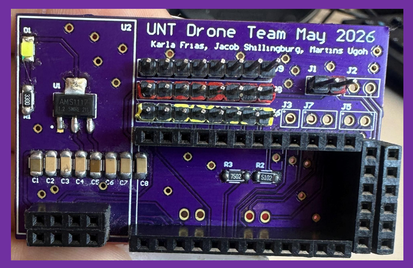

Abstract

This project presents the design and development of a head-tracking FPV camera control system for remote sensing application. The system allows an FPV camera to follow a user's head movements in real time, improving situational awareness and control for remotely operated platforms such as drones. The design integrates embedded control, wireless communication, servo actuation, and a custom PCB to create a compact and functional prototype. This work demonstrates the potential of head-tracking technology to support applications such as first responder assistance, surveillance, and remote navigation.

Introduction

In critical remote operations, every second and every viewing angle matters. Traditional FPV camera systems are limited because they often force the operator to rely on a fixed perspective, reducing awareness of the surrounding environment. This project addresses that challenge by developing a head-tracking camera system that translates natural head movement into real-time camera motion. The design integrates wireless communication, embedded control, servo motors, and a custom PCB to create a compact prototype that improves remote vision and operator control. By making camera movement more natural and intuitive, this system offers a practical step toward better remote sensing solutions for first responders in emergencies.

Methods

Electrical Design:

•Boards use an external power supply (Transmitter: 18650 3000mAh battery cell; Receiver: 3.7V LiPo Battery 500mAh;)

•On board linear regulator for 3.3V (AMS1117-3.3)

•Integrated filtering capacitors to improve antenna signal stability

•Processed head movement into control signals using MPU6050

•Wireless communication to send tracking data using NRF24L01 & E01-ML01DP5

•Utilized control signals for pan and tilt servos motion

•Designed an Arduino Pro Mini Carrier PCB in Altium

Mechanical Design:

•Designed and Printed 3D parts (Drone, Case, Gimbal)

•Assembled the camera and gimbal structure

PCB Design

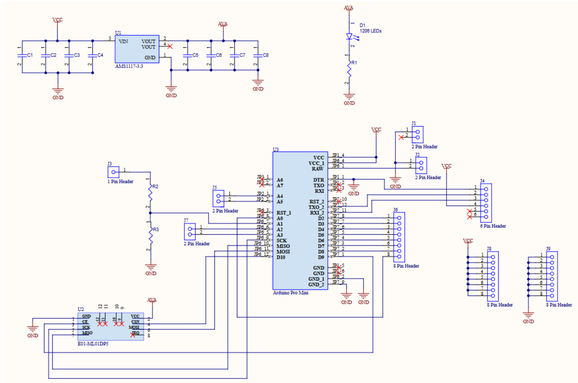

Figure 1. PCB Schematic in Altium

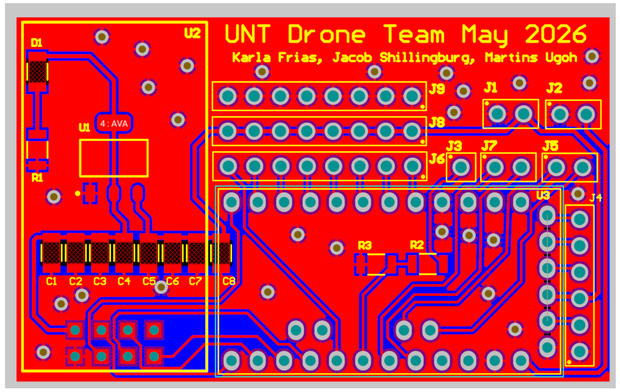

Figure 2. PCB Layout in Altium

Features:

2 layers

GND and 5V planes

Dimensions: 57mm*35mm

3D Design

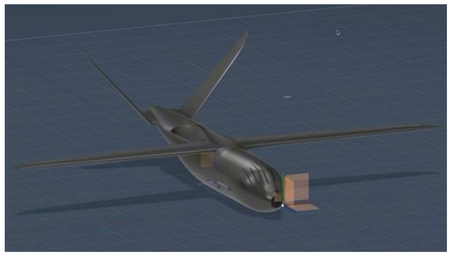

Figure 3. 3D CAD Design for our Drone in Fusion360

Figure 4. Printed 3D Design of Transmitter Case

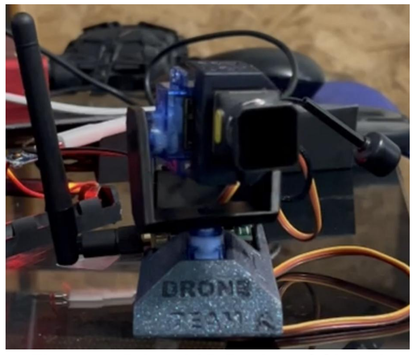

Figure 5. Printed 3D Design of Gimbal

Testing

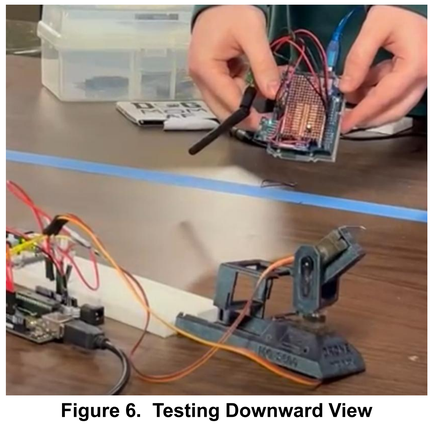

Figure 6. Testing Downward View

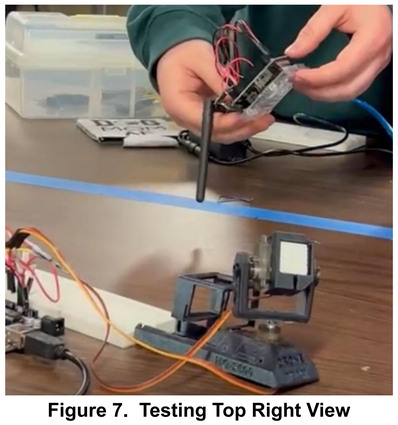

Figure 7. Testing Top Right View

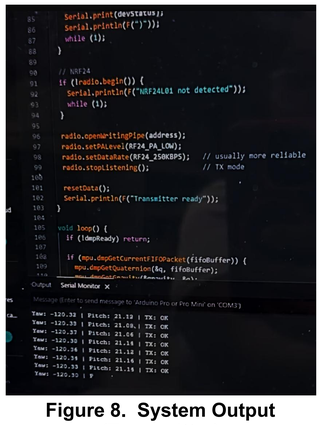

Figure 8. System Output (Transmitter)

Results

The final prototype successfully demonstrated real-time head-tracking camera control, showing that the overall system functioned as intended. Ground testing confirmed wireless communication over distances up to 1 km under testing conditions. During assembly and testing, the PCB supported system integration; however, the pin headers were prone to stripping, which created a reliability issue during repeated connections and adjustments. Although flight testing was not completed, the prototype verified the core functionality of the design and identified an area for future hardware improvement.

Conclusion

The developed head-tracking FPV camera system successfully demonstrated real-time camera movement based on user head motion. The project showed effective integration of electrical and mechanical design while improving remote viewing and operator awareness. This prototype provides a foundation for future use in drone and first responder applications.

Future Work

- Redesign connection points to reduce stripping issues
- Perform full flight testing
- Test Mechanical stability under drone movement
- Refine tracking response for smoother camera motion
- Update the PCB for improved durability (weather)
- Evaluate performance in field conditions
- Explore greater communication range and system upgrades

References

[1] MPU-6050, "Accelerometer/Gyroscope Sensor Module," documentation.

[2] AMS1117, "LDO Voltage Regulator," datasheet.

[3] WTO5 FPV camera, product documentation.

[4] MG90S micro servo motor, product documentation.

[5] E01-ML01DP5 transceiver module, documentation.

Acknowledgements

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